

The New Deal

GROUP PROJECT PROPOSAL





Challenge question

How might we build an **AI-powered tool** for patients with depression and anxiety based on methodologies of **Mindful Self Compassion** and **Internal Family Systems** to build adaptive journaling prompts for the patients based on individual needs and personality traits?

Depression & Anxiety



Benefit of technology

- ✧ **Accessibility and scalability**
- ✧ **Limitation of personalization**

→ Personalized Journaling App



Mindful Self Compassion

goal: to relate to oneself with greater kindness and understanding

- life satisfaction,
- decision making,
- self-esteem,
- emotional well-being,

Write a letter as if you were talking to a dearly beloved friend who was struggling with the same concerns as you.



IFS - evidence-based psychotherapy approach

Relationship between Self and the different parts of you:

Managers

Exiles

Firefighters

Question examples:

What memories from the past you can think of that cause you to feel negative emotions such as shame or anger?

In as much detail as possible describe that experience and try to identify what caused you to feel those emotions.



Personalization

Shortened Big 5 Personality Test:

Agreeableness



Openness



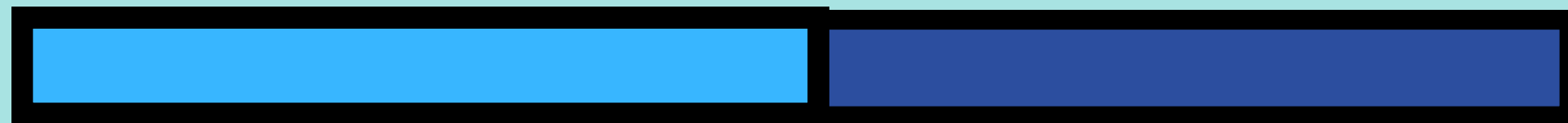
Conscientiousness



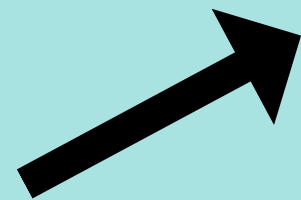
Neuroticism



Extraversion



Prompt Version 1



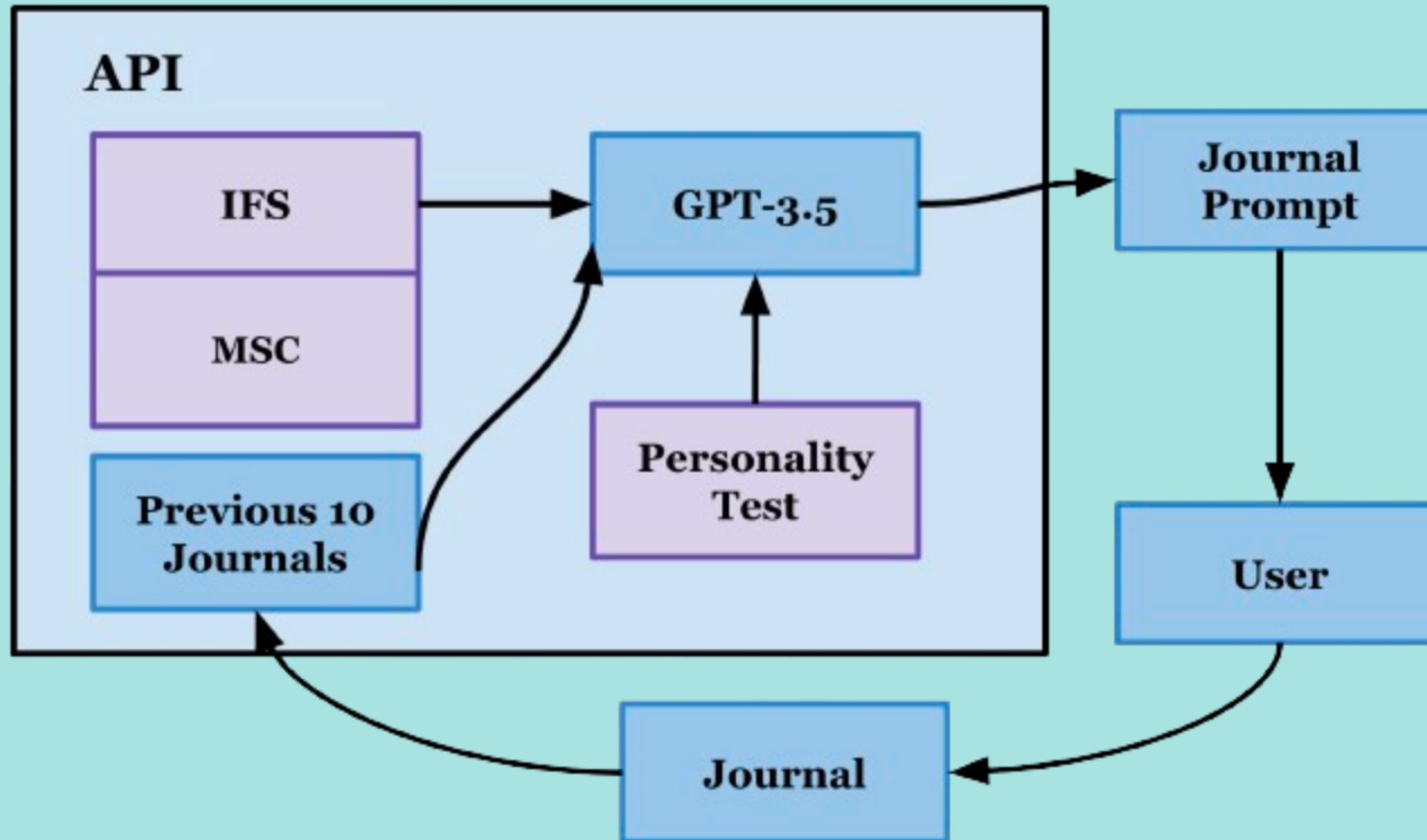
Prompt Version 2



How we will have it link in AI?

- The AI model will get the previous journals, personality type, and wellness test results as its input
- Then it will output a personalized prompt
- Patient will write a new journal from that prompt then take a wellness test to collect data on how effective it was
- Repeat

Flowchart for the App





Implementation (Tech Stack)

Frontend:

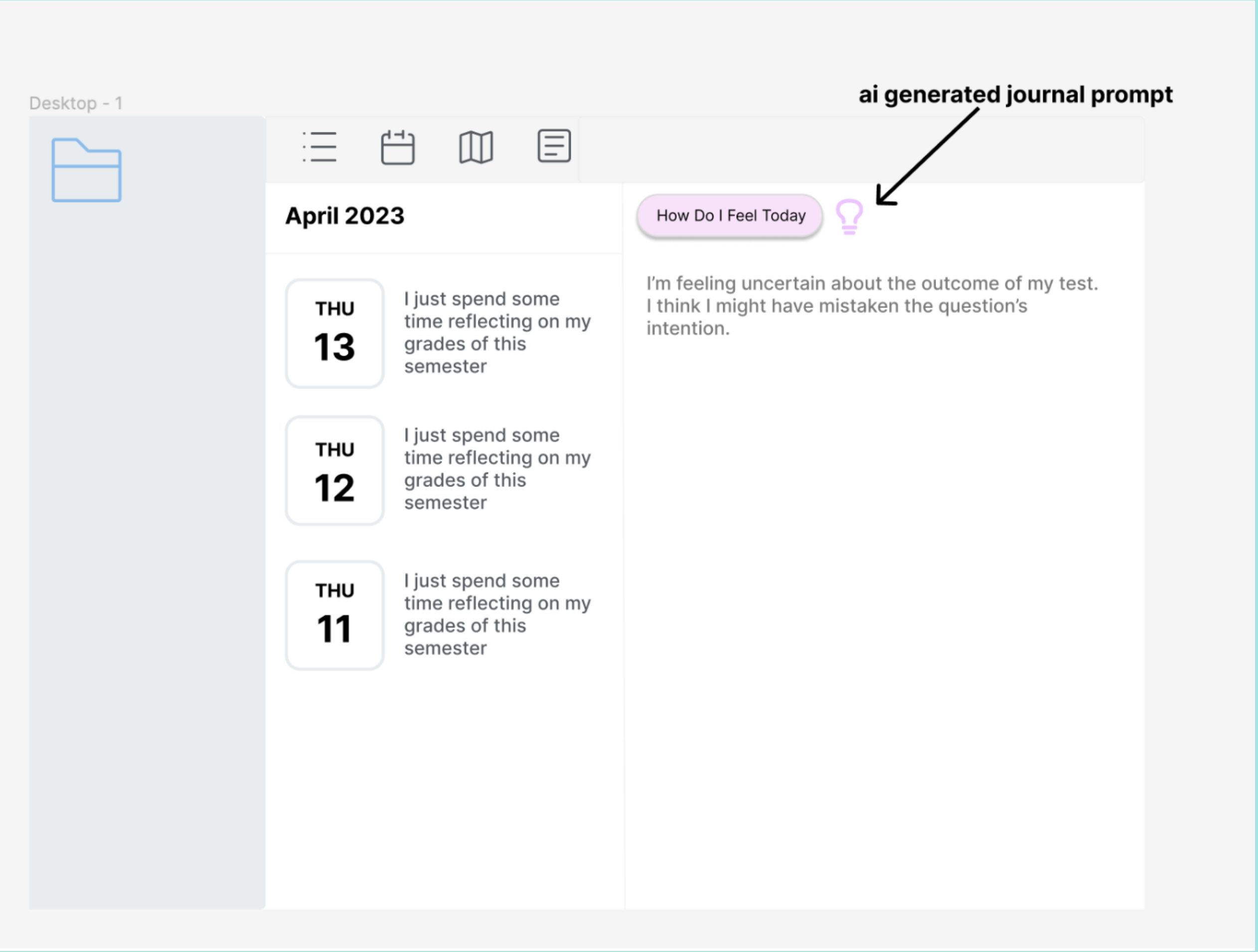
- Javascript
- WorkBox

Backend:

- Supabase
- OpenAI API

Visual Mockup

User takes
personality test and
report the results



Live Demo

Thank you!



References

Cherry, K. (2023, March 11). *Big 5 Personality Traits: The 5-Factor Model of Personality*. Verywell Mind. Retrieved December 10, 2023, from <https://www.verywellmind.com/the-big-five-personality-dimensions-2795422>

This is an article written by psychosocial rehabilitation specialist and author Kendra Cherry which has also been reviewed by a licensed clinical psychologist. Therefore, this was a reliable source to gain a basic understanding of the Big Five personality model and what each trait entails. Most project members have an understanding of the model through past education, however, a source was still used to ensure accuracy of understanding based off of a reliable source.

Internal Family Systems Therapy. (2022, May 20). Psychology Today. Retrieved December 12, 2023, from <https://www.psychologytoday.com/us/therapy-types/internal-family-systems-therapy>

The resources gives an overview of IFS model and its important parts that were described above. Additionally, the source provides with the examples of the mental health disorders that this model helps with.

Murray, B. (2002, June 1). *Writing to heal*. American Psychological Association. Retrieved December 10, 2023, from <https://www.apa.org/monitor/jun02/writing>

An article used to understand the potential benefits of journaling, which led to selection of a journaling application for the deliverable. This article was released by the American Psychological Association, a large and reputable body for psychology related knowledge. This source highlights the benefits of journaling including management of negative emotions and potential physical benefits such as faster healing.

Neff, K., & Germer, C. (2019). *The Transformative Effects of Mindful Self-Compassion*.

<https://mindfulwerkenmetcompassie.nl/wp-content/uploads/2020/03/The-Transformative-Effects-of-Mindful-Self-Compassion-Neff-Germer.pdf>

This sources provided us with examples of prompts that could have been used for Mindful Self-Compassion framework. It also explains Mindful Self-Compassion, and how it can be utilized to create the greatest results.

Phelan, H. (2018, October 25). *What's All This About Journaling?* The New York Times. Retrieved December 10, 2023, from <https://www.nytimes.com/2018/10/25/style/journaling-benefits.html>

New York Times article that is based off of an interview with a social psychologist who is a pioneer in writing therapy. The article gives a short summary of some of his studies, and how and why journaling is effective in aiding mental health. For example, how writing allows for labeling emotions and is a highly organizational process that helps organize one's thoughts. This source was used alongside the "Writing to heal" source when choosing to select a journaling application as the deliverable.